NQSync Quantum Simulator

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Bhopal

Student Declaration

We hereby declare that the work presented in the report entitled **"NQSync Quantum Simulator"** submitted by us for the completion of the summer internship in Quantum Network Simulation at Indian Institute of Science Education and Research, Bhopal, is an authentic record of our work carried out under the guidance of **Ankur Raina, QUCIS Lab** . Due acknowledgements have been given in the report for all material used. This work has not been submitted elsewhere for the reward of any other program.

Bhumika Upadhyay, IISER Bhopall & Aini Rizvi, VIT Bhopal University & Siddhant Jawalker, NSFU Bhopal **Place & Date:** October 8, 2025

Certificate

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Ankur Raina, QUCIS Lab **Place & Date:** October 8, 2025

Abstract

In the age where quantum technologies are emerging from the theory into the practice, the possibilities to investigate and play with quantum communication protocols should be the more abundant. We present NQSync, a quantum network simulation platform to support this need. With NQSync, users can visually simulate two popular quantum key distribution (QKD) protocols – Differential Phase Shift Quantum Key Distribution (DPS-QKD) and Coherent One-Way (COW) – in an intuitive drag-and-drop canvas interface. Rather than raw command-line simulators that require deep technical knowledge, NQSync makes quantum networking real with GUI interactivity. It allows users to build quantum networks with ease, to connect nodes, to set parameters, and to see protocol behaviors in real time. This tactile approach strips away the complexity surrounding quantum communication processes - NQSync is an essential teaching tool for researchers, teachers, and anyone interested in quantum communication. With scientific robustness merged with user friendly interface, NQSync intends to facilitate learning, experimentation, development in the dynamic world of quantum networks.

Acknowledgment

We Sincerely thank Dr. Ankur Raina, Assistant Professor, Department of Electrical Engineering & Computer Science, IISER Bhopal, for his invaluable advice and help. Dr. Raina's deep knowledge in quantum information theory, quantum cryptography, and quantum networks was invaluable in every stage of our effort on NQSync, spanning the conceptual design to the ultimate execution. His contribution improved not only our understanding of QKD protocols like DPS-QKD and COW but served as an inspiration for the vision of our useful drag-and-drop interface.

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Chapter 1

Introduction

1.1 Background

1.1.1 Quantum network simulation

Quantum Networks are practical implementaion of protocols that use many methods and practices of Quantum mechanics and quantum computing to transmit data from one node to other. Quantum networks are a rapidly developing area of communication technology, designed to deliver revolutionary new applications, including the distribution of cryptographic keys with provable security. The primary purpose of a quantum network is to generate, manage, and distribute cryptographic keys among various Quantum Key Distribution (QKD) nodes, providing these keys as a service to standard communication network users or applications. This is particularly crucial as advancements in quantum computing pose a threat to conventional computationally secure cryptographic methods like RSA and DH. [1]

[2] A quantum network typically incorporates both quantum channels and conventional (classical) channels that operate collaborative.[6g cow] The quantum channel is responsible for the sensitive work of QKD, exchanging quantum states on photons. These channels are often implemented using dark fibers [1]

There are many research works presented on working of Quantum Network Simulation and their practical implementations.

One such network is descirbed in Fig 1.1 shows a communication infrastructure called "Tokyo Qauntum Network" established across four districts and comprises of 6 stations. They discuss different implmentations of protocols such as BB84, COW QKD, DPS QKD etc. the figure below describes the network.

Different types of simulations of multiple protocols gives us the idea about the results and comparisions on real time.

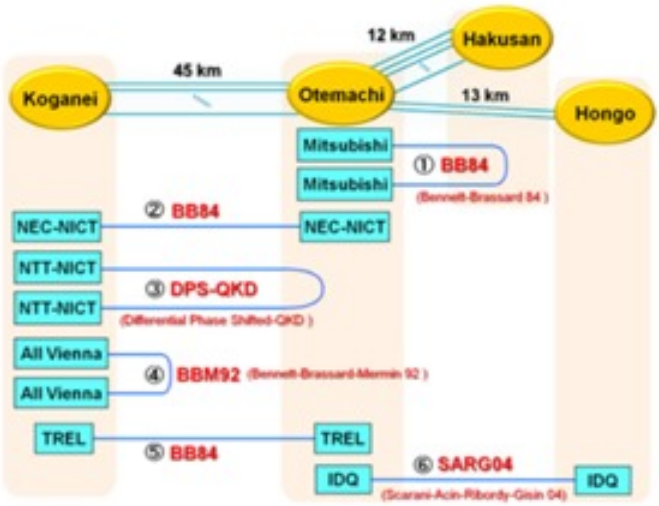


Figure 1.1: Tokyo Network

1.1.2 Simulations

SQUANCH

SQUANCH consist of a very basic architecture, it does not support multiprocessing capabilities but it works faster and supports simple error patterns.

In quantum computation, information is stored in states of qubits, this is the quantum generalization of a classical bit. A qubit in the physical world can be any of level quantum system, whether it is spin or polarization. A collection of such qubits form multipartical quantum systems. The states of a N-qubit quantum system lies on the 2N dimensional Hilbert space. It is similiar to classical algorithms which are represented as sequences of boolean logic gates, the quantum algorithms use quantum gates to manipulate the states of quantum systems. These algorithms transmit and manipulate the quantum information are frequently explained in terms of a colorful cast of characters starring Alice and Bob. In the SQUANCH framework, these "quantum actors" are represented by Agents. Conceptually,a quantum channel is a communication channel which transmits quantum information(qubits) and an associated noise model N which modifies the qubits, generally through coupling to the environment.

A typical workflow involves defining runtime logic for each agent, creating and formatting a QStream object, creating an output dictionary and agent instances, defining the network structure by connecting agents (e.g., alice.qconnect(bob) for quantum connections or alice.cconnect(bob) for classical connections), running the simulation, and finally, post-processing and visualizing the output data

SQUANCH can simulate various quantum phenomena and protocols, including quantum teleportation, superdense coding, man-in-the-middle attacks (where an eavesdropper attempts to intercept messages), and quantum error correction using models like the Shor Code to account for errors such as decoherence, depolarization, or qubit vanishing

1.1.3 COW Protocol

The Coherent One Way (COW) protocol works with encoding bit using pairs of empty and non-empty pulses.

COW protocol belongs to the class of QKD protocols referred to as DPR QKD protocols. In these kind of protocols, the security is derived from the coherence of sequential pulses.

- In the first step, Alice prepares a sequence of pulses $|0\rangle$—$\rangle$(empty, non-empty), —$\rangle$$|0\rangle$ (non-empty, empty) and —$\rangle$—$\rangle$ (non-empty, non-empty)
- While receiving the pulse sequence, Bob measures the time of arrival of 90% photons on his detector DB for the generation of raw key and rest 10% of photons on the monitoring line for security purpose
- Bob randomly checks the coherence between the non-empty pulses using detector DM1 and DM2 and the set up of Mach Zhender is in such a way that always DM1will click if there is no disturbance by Eve
- After that Alice and Bob perform sifting (removal of decoy sequences), error correction, privacy amplification to obtain the secret key.

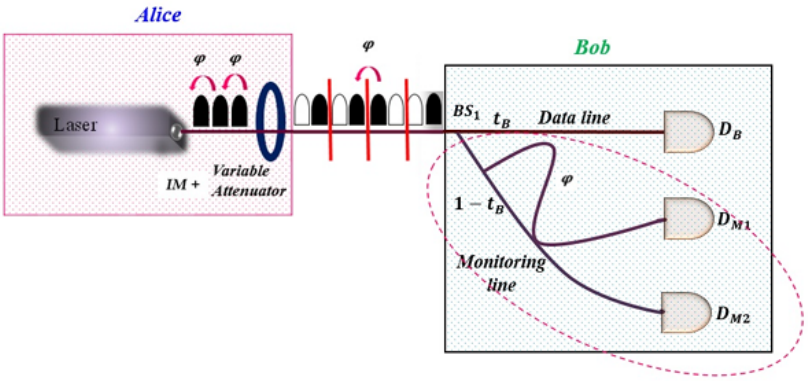


Figure 1.2: COW PRotocol

In one of the work by [4] experimental realization of an optical fiber based COW protocol for QKD in the telecom wavelength (1550 nm) where the attenuation in the optical fiber is minimum. The work explored the physical implmentation of cow protocol maintaining a qber rate below 6 percent.

The system has been developed to be rack mountable in two 19 inches boxes. Two fibres link them together; the first used for the quantum channel, the second for the classical communication, synchronisation and pre-sifting, which we explain momentarily. Each box is connected to the Internet for the classical processing of the raw key. Alice and Bob have similar structures: optical and electronics modules with Field Programmable Gate Arrays (FPGAs) and embedded computers. The optical modules are very simple (see Figure 1.2).

1.1.4 DPS QKD Protocol

The DPS QKD protocol was proposed in 2002 by Inoue et al. [19]. In this protocol, Alice creates a weak coherent pulse (WCP) whose average photon number is 0.2 and randomly modulates the phase either by 0 or using a phase modulator. She sends these modulated pulses to Bob using an optical fiber. At the receiver's side, Bob passes the received pulses to a 1 bit delay Mach Zehnder interferometer. The outputs of the Mach Zehnder interferometer are measured either by single photon detectors (SPDs) or superconducting nanowire single photon detectors (SNSPDs). The two consecutive pulses interfere with each other in the Mach Zehnder interferometer with the detection from either of the two detectors having information of the phase difference between between the two consecutive pulses. In Fig. 2, click on DM1 (DM2) means phase difference 0 () between the two successive pulses.

As descibed in [5] the DPS QKD work as follows

- Alice sends a random sequence of phase-modulated (0,) attenuated pulses to Bob, with each pulse having an average photon number less than one.
- Bob uses photon detectors and a 1 bit delay Mach Zehnder interferometer to measure the phase difference between two consecutive pulses and records the information of detector clicks as well as time of arrival of photons.
- Bob publicly informs Alice about the times when either of the detectors clicked
- With the timing information about Bob's detectors clicks, Alice obtains 'which detector clicks' information at Bob's side.
- Key bits 0 and 1 are respectively assigned if the phase difference between two successive pulses are 0 or .
- Alice and Bob perform error correction and privacy amplification on the sifted key to obtain the secret key.[5]

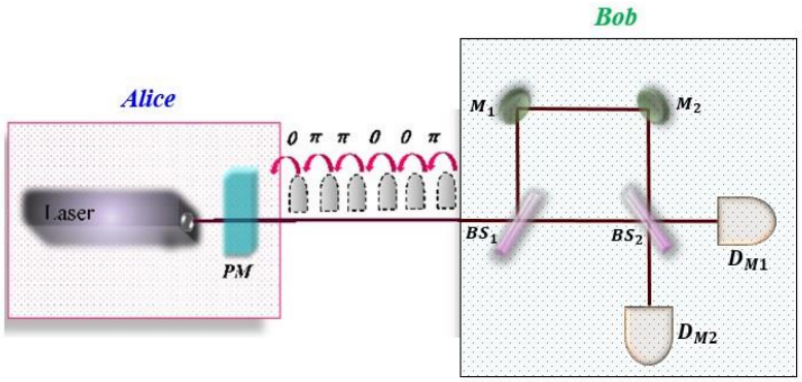


Figure 1.3: DPS QKD protocol

Chapter 2

Literature Review

[1]

2.0.1 High-Fidelity Coherent-One-Way QKD Simulation Framework for 6G Networks: Bridging Theory and Reality

The paper introduces Coherent one Way Quantum Key distribution (COW QKD) simulation framework, developed to simulate practical implementation challenges for future 6g networks. Real-life implemenations of such QKD networks are expensive, Simulators on the other hand gives flexibilty of running multiple experiments without the use of equipments. QKD offers information-theoretic security, meaning it guarantees unbreakable encryption by relying on quantum mechanics principles, such as the detection of eavesdropping attempts through system disturbance.

The paper explains the elements of QKD systems and applications of COW QKD protocols which have low-complexity and high performance in respect to Quantum Bit error rates and secreacy key rates. they conducted extensive experiments across various transmission distances (1 km, 2 km, 3.5 km, and up to 14.4 dB/km fiber loss), both with and without simulated eavesdropping. The results demonstrated an acceptable match between the simulations and the experimental findings for both COW3 and COW4 protocols, confirming the simulation framework's suitability as a tool for designing QKD communication systems. Furthermore, the study provided crucial design guidelines, illustrating the impact of factors like transmission power, fiber loss, eavesdropping scale, and fiber length on QBER, emphasizing the importance of low fiber loss and short fiber lengths for maintaining low error rates, even in the presence of eavesdropping.

2.0.2 QUANTUM NETWORK

The paper gives a overview of quantum networks and their potential of providing different applications with security. Quantum network simulations are vital for undertsanding the comparision between different network's architecture. The Tokyo Quantum Network is presented as a practical example, which spans four districts in Tokyo and features six stations, integrating a classical

network with quantum channels to allow all stations to communicate securely. This network employs various QKD protocols, including the simpler BB84 protocol (which encodes keys in photon polarization, relying on the no-cloning theorem for security) and more complex ones like Differential Phase Shift Quantum Key Distribution (DPS-QKD), which uses randomly phase-modulated coherent pulse trains.

This document provides an overview of quantum networks, emphasizing their potential to deliver revolutionary applications, including the distribution of cryptographic keys with provable security. It highlights how quantum network simulations are vital for understanding trade-offs in different network architectures, optimizing quantum hardware, and developing robust control planes. The Tokyo Quantum Network is presented as a practical example, which spans four districts in Tokyo and features six stations, integrating a classical network with quantum channels to allow all stations to communicate securely. This network employs various QKD protocols, including the simpler BB84 protocol (which encodes keys in photon polarization, relying on the no-cloning theorem for security) and more complex ones like Differential Phase Shift Quantum Key Distribution (DPS-QKD), which uses randomly phase-modulated coherent pulse trains. The document also delves into several software simulation tools that facilitate the development and understanding of quantum networks. SQUANCH is highlighted as a user-friendly open-source simulator, noted for its simplicity and ability to support various error patterns, making it suitable for realistic simulations. SQUANCH's framework represents "quantum actors" as Agents and allows for the simulation of complex quantum communication scenarios such as quantum teleportation (transferring quantum states using classical communication and entangled pairs), superdense coding (sending two classical bits with a single qubit via an entangled pair), and even man-in-the-middle attacks, along with quantum error correction using codes like Shor Code. Additionally, QKDNetSim is introduced as a module that incorporates major functional elements of QKD networks, including quantum and classical layers, key management, and post-processing applications, and Qiskit is mentioned as a comprehensive software collection for executing and optimizing quantum circuits on quantum computers

2.0.3 Implementation of coherent one way protocol for quantum key distribution up to an effective distance of 145 km

[4] This paper details the experimental realization of an optical fiber-based Coherent One-Way (COW) QKD protocol at the telecom wavelength of 1550 nm, achieving an effective distance of up to 145 km. The primary motivation for this work is to provide unconditional security against the threat of scalable quantum computers, which are expected to render traditional cryptographic schemes (like RSA and Diffie-Hellman) vulnerable. COW QKD is highlighted as a practical and secure protocol due to its ease of experimental implementation, robustness against photon number splitting (PNS) attacks, tolerance against reduced visibility, and polarization insensitivity, making it suitable for fiber-based communications without complex polarization control devices. The experiment used a laser with a 500 MHz repetition rate, generating pulses with an average photon number of 0.5, and employed a 90:10 optical coupler to distribute photons between the data line (for raw key generation) and a monitoring line (for security checks).

The study rigorously analyzed the impact of various experimental parameters on the final key rate (KR), including disclose rate (DR), compression ratio (CR), detector dead time (DT), and excess bias voltage (BV), while consistently maintaining the Quantum Bit Error Rate (QBER) below 6%. It was observed that KR linearly increases with decreasing DR and CR, and generally increases with decreasing DT, though abrupt changes in KR were noted for certain DT values at longer distances due to increased dark count rates. KR was found to remain stable for various values of BV and over time. The achieved key rates, ranging from 154 bps to 14562 bps depending on distance and parameter choices, are comparable to or better than other state-of-the-art COW QKD implementations, especially considering the use of standard telecom-grade optical fiber and the absence of trusted nodes for similar distances.

2.0.4 Experimental implementation of distributed phase reference quantum key distribution protocols

[5] This paper focuses on the experimental realization of Distributed Phase Reference (DPR) Quantum Key Distribution (QKD) protocols, specifically Coherent One-Way (COW) and Differential Phase Shift (DPS) QKD, utilizing optical fiber at telecom wavelengths. DPR protocols are characterized by encoding information in the phase difference between sequential pulses or photon arrival times, using weak coherent pulses rather than single photons, which makes them highly practical and easier to implement with existing technology compared to other QKD schemes. The research aims to comprehensively analyze the key rates (KR) of these protocols based on various parameters, including distance, disclose rate (DR), compression ratio (CR), and detector dead time (DT), addressing a gap in previous studies that often overlook these critical factors.

The study implemented DPS QKD over distances of 80 km, 100 km, and 120 km, and a COW QKD setup that now includes a monitoring line, improving upon prior work. Experimental results for DPS QKD consistently showed that KR increases with a decrease in both DR and CR, and also with a decrease in detector DT. A significant finding from the comparative analysis is that DPS QKD achieved substantially higher key rates than COW QKD under similar experimental conditions (e.g., 21352 bps for 80 km DPS vs. 2000-4000 bps for 100 km COW with similar DT and CR settings). This difference is largely attributed to the higher repetition rate (1 GHz for DPS vs. 500 MHz for COW). The paper emphasizes that the detailed analysis of how DR, CR, and DT influence KR provides valuable insights for optimizing QKD system performance and progressing towards secure network realization

Chapter 3

Methodology

3.1 Simulator architecture

The fundamental framework of a QKD communication process incorporates both quantum and conventional channels that operate collaboratively. The quantum channel is responsible for the sensitive work of QKD, enabling security through the unique physical properties of quantum information carriers.The conventional channel supports essential post-processing sessions, including key agreement and eavesdropper detection between Alice and Bob. This dual-channel structure is crucial for ensuring that unauthorized access is quickly identified and eliminated.

The COW protocol relies mostly on passive optical components, contributing to its simplicity of implementation. It is also polarization insensitive, making it suitable for fiber-based communications without complex polarization control devices. The system typically involves Alice (the sender), a Quantum Channel, and Bob (the legitimate receiver), alongside a Conventional Channel

In DPS QKD, Alice creates a weak coherent pulse (WCP) with an average photon number of 0.2 and randomly modulates the phase (0 or) using a phase modulator. She sends these modulated pulses to Bob via an optical fiber. At the receiver's side, Bob passes the received pulses through a 1-bit delay Mach-Zehnder interferometer

The web based solution for this quantum network simulation inlcudes components following the flowchart:

3.2 Paramters

3.2.1 Common Parameters for COW & DPS QKD

Both COW and DPS QKD protocols use certain parameters to simulate the network. In this section we discuss about the parameters that are common in both protocols and how they impact the network results.

- distance - Represents the distance of a fiber channel in kilometer. The more is the distance

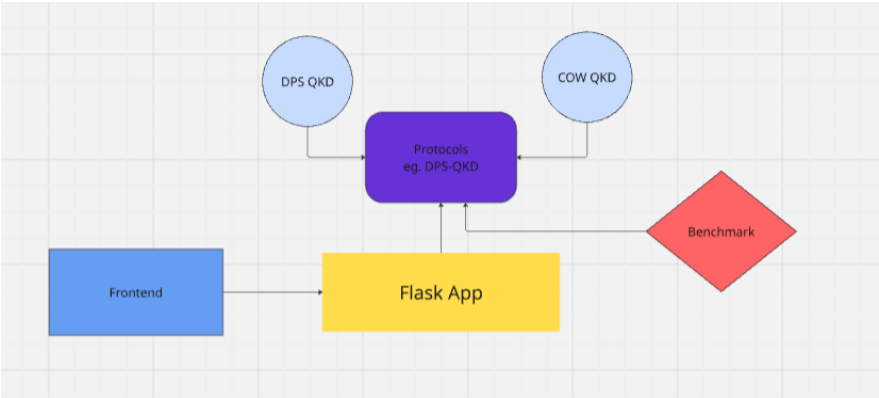


Figure 3.1: FLowchart

more is the photon loss and transmittance decreases exponentially. It is a user dependent value. The protocol is set to work between 50 to 100km.

- fiber-type - There are two types of fiber-type 'underground' and 'aerial'. It depends on the input user sets. Aerial fiber have higher loss as the alpha values associated with 'aerial' is 0.30 and 'underground' is 0.20.
- mu - Is the average photons per pulse. If mu values are set higher there will be more detections but also increase in errors.
- pulse-rate - Is the laser pulse rate, if rate is higher there is a increase in raw key bits but they are limited by the dead-time parameter.
- det-eff - Represents the detector efficiency, higher detector efficiency means more pulse detections.
- dark-rate - Number of dark counts in one run, if the dark counts is set higher more noise will be produced in the detection.
- dead-time - Represents the detector recovery time in sec, longer is the dead time lower is the detection rate which impacts the raw bit count.
- visibility - Shows how the interference visibility is in the protocol, lower is the visibility higher is the QBER.

3.2.2 COW Protocol Specific Parameters

Some paramters used in the simulation are COW protocol specific and are only used to simulate the results of the same protocol. Below are the COW specific protocols.

- error-correction-efficiency - It represents how efficiently errors are corrected. The more is the parameter better is the final key rate.

- decoy-ratio - This is a fixed value set at 0.30, it represents the fraction of pulses that uses decoy states, it may be used further to detect eavesdropping while stimulating the protocols.
- mu-signal & mu-decoy - shows intensities of signal and decoy states which also affects the security and detection rates.

3.2.3 DPS Protocol Specific Parameters

These parameters are only applied to simulate and get results from DPS QKD protocol.

- discosure-rate - Fractions of bits that are revealed for error checking, the higher are the rate the lower is the final key rate.
- compression-ratio - It is the fraction of bits that is kept after privacy amplifications. Lower the ratio more secure the protocol is, but if ratio is set to be lower it decreases the count of the bits.

3.3 Protocols

3.3.1 COW Protocol

In this section we discuss the architecture of the simualtion of the COW QKD protocol.

Alice Transmitter

In the Alice setup, the transmitter generates weak coherent laser pulses with mean photon number mu. It randomly inserts 30 percent decoy pulses into the raw signal. Then we pair them between 0,1 and decoy depeneding on the presence and abcense of pulse pairs.

Quantum Channel

It is the simulation of the optical fiber with the 'attenuation coefficient' set at 0.2 for underground and 0.3 for aerial.

Bob's Reciever

In the bob setup the receiver detectes the pairs of pulses and uses time resolved detection to distinguish between the signal and decoy states. the basis are detected through random phase modulation.

Channel Loss

The transmittance is calculated for fiber attenuation by:

$$\eta = 10^{-\alpha \times \text{distance}_{\text{km}}/10}$$

Detection Probabilities

$$p_{\text{signal}} = \mu \times \eta \times \text{det}_{\text{eff}}$$

$$p_{\text{dark}} = \frac{\text{dark}_{\text{rate}}}{\text{pulse}_{\text{rate}}}$$

Decoy State Analysis

The signal intensity is set at mu and decoy depends 0.1 times the signal density.

QBER Calculation

$$\text{min}_{\text{qber}} = \max(0.005, 0.005 + 0.02 \times \text{distance}_{\text{km}}/100)$$

$$\text{effective}_{\text{visibility}} = 1 - \text{min}_{\text{qber}}$$

Key Rate Derivation

$$R_{\text{raw}} = \frac{\text{pulse}_{\text{rate}} \times p_{\text{signal}}}{1 + p_{\text{signal}} \times \text{pulse}_{\text{rate}} \times \text{dead}_{\text{time}}}$$

$$R_{\text{sifted}} = 0.5 \times R_{\text{raw}}$$

$$R_{\text{final}} = R_{\text{sifted}} \times (1 - \text{error}_{\text{correction_efficiency}} \times \text{QBER})$$

3.3.2 DPS QKD Protocol

DPS (Differential Phase Shift) QKD encodes information in the phase difference between consecutive light pulses rather than individual pulses. It's efficient and robust against certain attacks. DPS encodes data in phase changes between pulses, uses interference to decode, and achieves security through phase sensitivity and privacy amplification. The web simulator lets you tweak physical parameters to see their impact on performance.

Alice transmitter

Alice generates a series of weak laser pulses with carefully controlled phases. Each bit is encoded by either maintaining the phase between consecutive pulses (representing '0') or introducing a phase shift (representing '1'). The intensity of these pulses is kept low (typically ¡ 1 photon per pulse) to maintain quantum security.

Quantum channel

The encoded pulses travel through an optical fiber, experiencing attenuation that increases with distance. The fiber's loss characteristics (typically 0.2 dB/km for underground or 0.3 dB/km for aerial) significantly impact the number of photons reaching Bob. Environmental factors like temperature fluctuations can introduce additional phase noise.

Bob's Receiver

Bob uses a 1-bit delay interferometer to measure phase differences between successive pulses. This interferometer creates interference between each pulse and its immediate predecessor. Two single-photon detectors monitor the output ports, allowing Bob to decode the phase-encoded information based on which detector registers photons.

Phase Encoding

Alice encodes bits as:

$$\text{phase} = \pi \times \text{bit}_{\text{value}} \quad (0 \text{ or } \pi)$$

Interference Measurement

Bob detects phase difference between pulses n and n-1:

$$\text{phase_diff} = (\text{phase}[n] - \text{phase}[n-1]) \bmod 2\pi$$

$$\text{bit} = \begin{cases} 0 & \text{if phase_diff} \approx 0 \\ 1 & \text{otherwise} \end{cases}$$

Error Model

$$\text{error_prob} = \text{loss}_{\text{dB}} \times 0.0085 \quad \text{(Empirical relationship)}$$

Key Rate Derivation

[Raw detection rate (same as COW)

Sifted rate (50)

Final rate after privacy amplification:

$$R_{\text{final}} = R_{\text{sifted}} \times (1 - \text{disclosure_rate}) \times (1 - \text{compression_ratio})$$

Frontend

The client-side implementation consists of multiple interactive modules built with modern web technologies:

- Dynamic Form Interfaces: Protocol-specific input forms with real-time validation

  Range-limited sliders for physical parameters (distance, , etc.)

  Tooltip explanations of QKD terminology

- Visualization Dashboard: Live-updating graphs of key rate vs. distance

  QBER heatmaps showing error probability landscapes

  Animated photon transmission diagrams

- Network Topology Canvas: Drag-and-drop quantum node placement

  Interactive fiber path drawing with loss calculations

  Protocol configuration panels for each quantum link

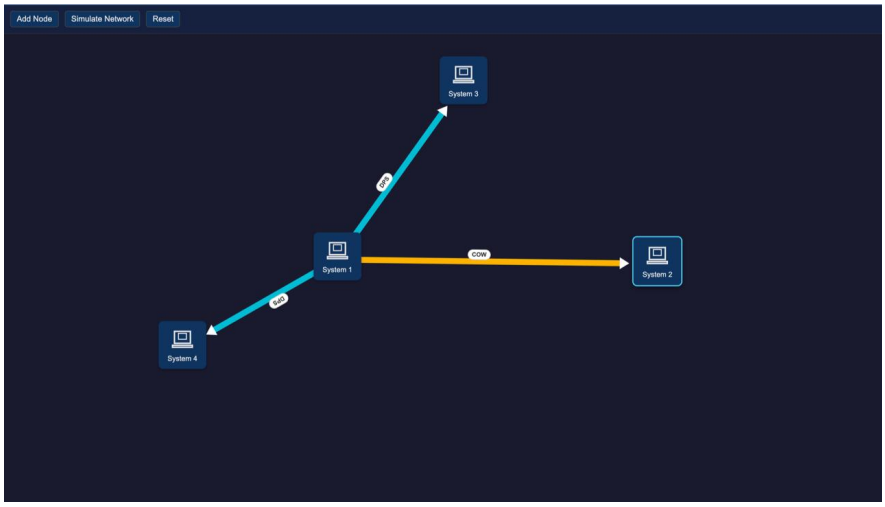


Figure 3.2: Canvas

Backend

The backend provides two primary simulation functions: `simulate_cow_qkd()` for the Coherent One-Way protocol and `simulate_dps_qkd()` for Differential Phase Shift QKD. Each function accepts multiple physical parameters, including transmission distance, fiber type, and detector characteristics, and returns comprehensive results such as key rates, quantum bit error rates (QBER), and detection statistics.

The web interface is implemented through several routes: form handlers (`/cow-form`, `/simulator-form`) for user input, simulation endpoints (`/save-cow-simulation`, `/save-simulation`) for processing requests, and data retrieval endpoints (`/get-results`, `/results`) for accessing simulation outputs. Input validation is handled by the `safe_float()` utility function, which sanitizes user inputs before processing. Basic CRUD operations are supported through endpoints such as `/remove-link` and `/reset-all`.

The system employs in-memory storage without database persistence, prioritizing simplicity and core simulation functionality. Advanced features such as user authentication and asynchronous task processing are not currently implemented, allowing the system to focus on delivering QKD simulations and web-based result visualization. [3, 1].

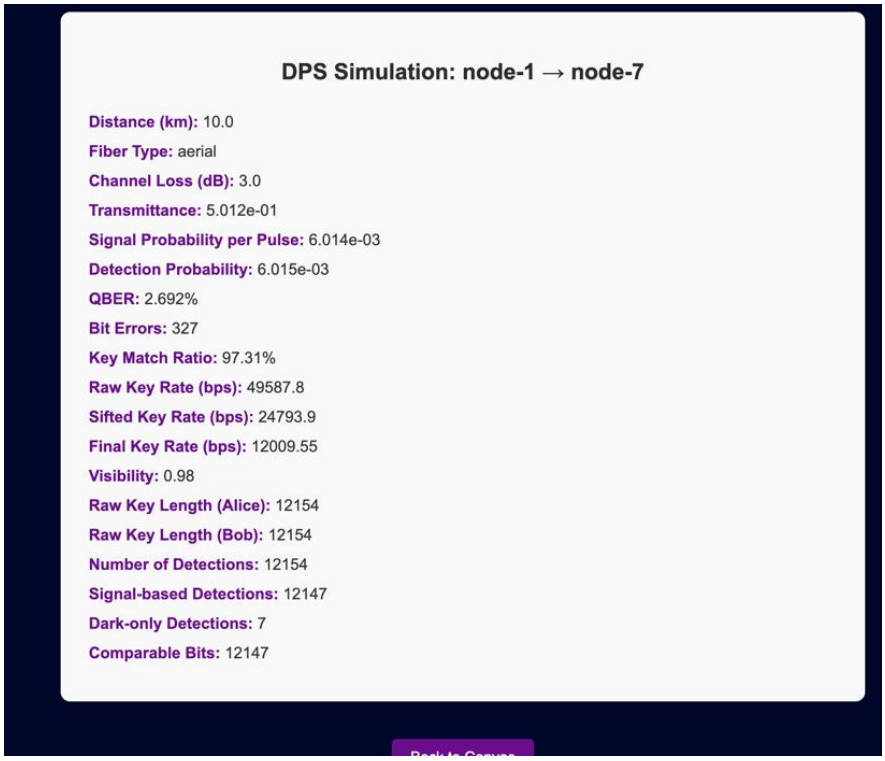


Figure 3.3: Result

Chapter 4

Results

4.1 COW QKD Results

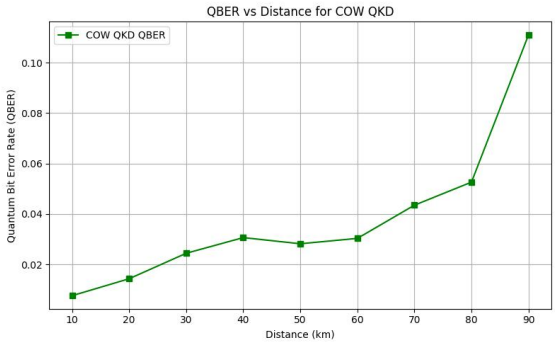


Figure 4.1: Underground view

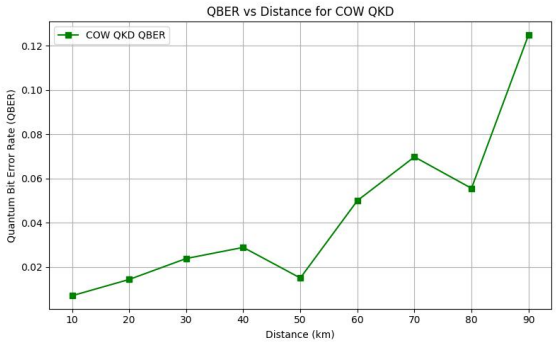


Figure 4.2: Aerial view

4.2 DPS QKD Results

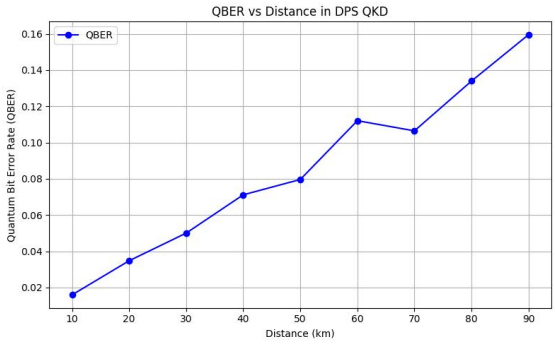


Figure 4.3: Underground view

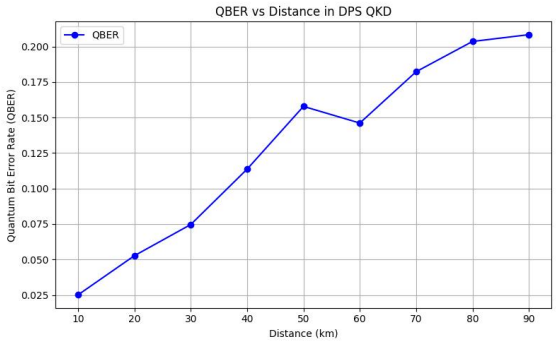


Figure 4.4: Aerial view

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